SE LAB 05 – OUTPUT SCREENSHOTS

Pre-Modification

Pylint on the original code (5.53 / 10)

```
D:\Projects\SE_06_it_helpdesk>type reports\before\pylint_before.txt
************* Module helpdesk_system
helpdesk_system.py:13:0: C0301: Line too long (118/100) (line-too-long)
helpdesk_system.py:1:0: C0114: Missing module docstring (missing-module-docstring)
helpdesk_system.py:7:0: C0103: Constant name "ticket_counter" doesn't conform to UPPER_CASE naming style (invalid-name)
helpdesk_system.py:10:0: C0116: Missing function or method docstring (missing-function-docstring)
helpdesk_system.py:10:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
helpdesk_system.py:11:4: W0603: Using the global statement (global-statement)
helpdesk_system.py:14:23: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
helpdesk_system.py:17:0: C0116: Missing function or method docstring (missing-function-docstring)
helpdesk_system.py:20:0: C0116: Missing function or method docstring (missing-function-docstring)
helpdesk_system.py:23:4: W0702: No exception type(s) specified (bare-except)
helpdesk_system.py:26:0: C0116: Missing function or method docstring (missing-function-docstring)
helpdesk_system.py:30:0: C0116: Missing function or method docstring (missing-function-docstring)
helpdesk_system.py:30:0: C0103: Function name "unresolvedHighPriority" doesn't conform to snake_case naming style (invalid-name)
helpdesk_system.py:32:4: C0206: Consider iterating with .items() (consider-using-dict-items)
helpdesk_system.py:33:43: C0121: Comparison 'tickets[t]['tech'] == None' should be 'tickets[t]['tech'] is None' (singleton-comparison)
helpdesk_system.py:37:0: C0116: Missing function or method docstring (missing-function-docstring)
helpdesk_system.py:40:0: C0116: Missing function or method docstring (missing-function-docstring)
helpdesk_system.py:41:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
helpdesk_system.py:41:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
helpdesk_system.py:45:0: C0116: Missing function or method docstring (missing-function-docstring)
helpdesk_system.py:47:4: W0612: Unused variable 't2' (unused-variable)

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Your code has been rated at 5.53/10
```

Bandit on the original code (1 High, 3 Low)

```
D:\Projects\SE_06_it_helpdesk>type reports\before\bandit_before.txt
.\venv\Scripts\python.exe : [main]      INFO    profile include tests: None
At line:1 char:1
+ .\venv\Scripts\python.exe -m bandit helpdesk_system.py *> bandit_repo ...
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : NotSpecified: ([main]     INFO    profile include tests: None:String) [], RemoteException
    + FullyQualifiedErrorId : NativeCommandError

[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.12.10
Run started:2026-10-16 04:00:18.883118+00:00

Test results:
>> Issue: [B404:blacklist] Consider possible security implications associated with the subprocess module.
   Severity: Low   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_imports.html#b404-import-subprocess
   Location: .\helpdesk_system.py:2:0
1       import json
2       import subprocess
3       from datetime import datetime

--------------------------------------------------
>> Issue: [B105:hardcoded_password_string] Possible hardcoded password: 'helpdesk@123'
   Severity: Low   Confidence: Medium
   CWE: CWE-259 (https://cwe.mitre.org/data/definitions/259.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b105_hardcoded_password_string.html
   Location: .\helpdesk_system.py:8:17
7       ticket_counter = 100
8       ADMIN_PASSWORD = "helpdesk@123"
9
```

```
--------------------------------------------------
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b110_try_except_pass.html
   Location: .\helpdesk_system.py:23:4
22              tickets[ticket_id]["notes"].append(note)
23          except:
24              pass
25

--------------------------------------------------
>> Issue: [B602:subprocess_popen_with_shell_equals_true] subprocess call with shell=True identified, security issue.
   Severity: High   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b602_subprocess_popen_with_shell_equals_true.html
   Location: .\helpdesk_system.py:38:11
37      def ping_user_machine(hostname):
38          return subprocess.call("ping -c 1 " + hostname, shell=True)
39

--------------------------------------------------

Code scanned:
        Total lines of code: 47
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 3
                Medium: 0
                High: 1
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 1
                High: 3
Files skipped (0):
```

Flake8 on the original code

```
D:\Projects\SE_06_it_helpdesk>type reports\before\flake8_before.txt
original\helpdesk_system_original.py:10:1: E302 expected 2 blank lines, found 1
original\helpdesk_system_original.py:13:80: E501 line too long (118 > 79 characters)
original\helpdesk_system_original.py:14:80: E501 line too long (86 > 79 characters)
original\helpdesk_system_original.py:17:1: E302 expected 2 blank lines, found 1
original\helpdesk_system_original.py:20:1: E302 expected 2 blank lines, found 1
original\helpdesk_system_original.py:23:5: E722 do not use bare 'except'
original\helpdesk_system_original.py:26:1: E302 expected 2 blank lines, found 1
original\helpdesk_system_original.py:30:1: E302 expected 2 blank lines, found 1
original\helpdesk_system_original.py:33:63: E711 comparison to None should be 'if cond is None:'
original\helpdesk_system_original.py:37:1: E302 expected 2 blank lines, found 1
original\helpdesk_system_original.py:40:1: E302 expected 2 blank lines, found 1
original\helpdesk_system_original.py:45:1: E302 expected 2 blank lines, found 1
original\helpdesk_system_original.py:47:5: F841 local variable 't2' is assigned to but never used
original\helpdesk_system_original.py:52:80: E501 line too long (80 > 79 characters)
original\helpdesk_system_original.py:58:1: E305 expected 2 blank lines after class or function definition, found 1
```

Post-Modification

Pylint on the final code (10.00 / 10)

```
D:\Projects\SE_06_it_helpdesk>python -m pylint helpdesk_system.py

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Your code has been rated at 10.00/10
```

Bandit on the final code (0 High, 0 Medium, 2 Low)

```
D:\Projects\SE_06_it_helpdesk>python -m bandit -r helpdesk_system.py
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.12.10
Run started:2026-10-16 10:26:15.702229+00:00

Test results:
>> Issue: [B404:blacklist] Consider possible security implications associated with the subprocess module.
   Severity: Low   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_imports.html#b404-import-subprocess
   Location: .\helpdesk_system.py:7:0
6       import re
7       import subprocess
8       from datetime import datetime

--------------------------------------------------
>> Issue: [B603:subprocess_without_shell_equals_true] subprocess call - check for execution of untrusted input.
   Severity: Low   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b603_subprocess_without_shell_equals_true.html
   Location: .\helpdesk_system.py:126:17
125         try:
126             result = subprocess.run(
127                 command,
128                 check=False,
129                 timeout=10,
130                 capture_output=True,
131                 text=True,
132             )
133         except (subprocess.TimeoutExpired, OSError):

--------------------------------------------------

Code scanned:
        Total lines of code: 136
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 2
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 2
Files skipped (0):
```

Flake8 on the final code (no violations)

```
D:\Projects\SE_06_it_helpdesk>python -m flake8 helpdesk_system.py

D:\Projects\SE_06_it_helpdesk>
```